***Tutorial for geochemical and reactive transport modelling using the HPx framework – Introducing biogeochemical reaction modelling using PHREEQC***

# CONTENTS


Contents ..... 4
List of Figures ..... 6
List of Tables ..... 7
List of Code Extracts ..... 8
Glossary ..... 9
Abbreviations ..... 10
PREFACE ..... 12
Series volumes ..... 12
1. Introduction ..... 1
2. Notes on chemistry of cement systems ..... 2
3. Description of the cement module ..... 4
3.1 Download and install ..... 4
3.2 Templates ..... 5
4. Thermodynamic modelling of cement hydrates ..... 7
4.1 Set-up HPGeochemistry ..... 8
4.2 Modelling CEM-I hydrate assemblage and pore water composition ..... 10
4.2.1 Calculation of the hydrate assemblage for a CEM-I oxide composition ..... 10
4.2.2 Limited replacement OPC with GBFS ..... 14
4.2.3 Reaction path model with varying w/c ratio ..... 17
4.2.4 Reaction path model with varying temperature ..... 17
4.3 Modelling CEM-II and CEM-III hydrated phases and pore water composition ..... 18
4.3.1 Reaction path model with replacements ..... 18
5. Cement hydration ..... 23
6. Modelling degradation processes – Thermodynamic approaches (mixing tank approaches) ..... 27
6.1 Ingress ..... 28
6.2 Exchange ..... 31
6.3 Leaching ..... 32
7. Modelling degradation processes – Reactive transport at pore scale ..... 36
8. Modelling degradation processes – Reactive transport at continuum scale ..... 37
9. Modelling degradation processes – Surrogate modelling ..... 38
Appendix A Input and output files ..... 41
9.1 Input tables ..... 41
9.1.1 Output (blending, ingress, leaching) ..... 41
9.2 ..... 41
9.2.1 Set-up a template simulation ..... 41

References ..... 42
Index ..... 43

# GLOSSARY

| | |
|---|---|
| Case | When a simulation contains an input table and parameter values are defined in that table, a case represents one particular calculation in the simulation with that set of parameters. Calculation results can be saved in a separate subfolder in the simulation folder.<br>*See*<br>Tutorial 4-3 Define cases in a simulation |
| Parametric study | Based on a global input table, a series of calculations will be performed in which selected numerical values will vary automatically based on (i) start or end position of a parameter vector, (ii) parameter vectors listed in a table, (iii) centered around a parameter vector, or (iv) gridded parameter vectors.<br>*See*<br>Tutorial 4-5 |

# 3. DESCRIPTION OF THE CEMENT MODULE

## 3.1 Download and install

The cement module is available on github via this link. One need to download a zip file, unzip it and bring the module .

A module should be stored in the general Module directory of HPx (HYDRUS 5.x) and can be used in several projects. Tutorial 3-1 shows how to obtain the module and place it in the correct folder such that it can be activated in your projects.

*TUTORIAL 3-1 INSTALL CEMENT MODULE*

**Description**

To add the cement module to HPGeochemistry

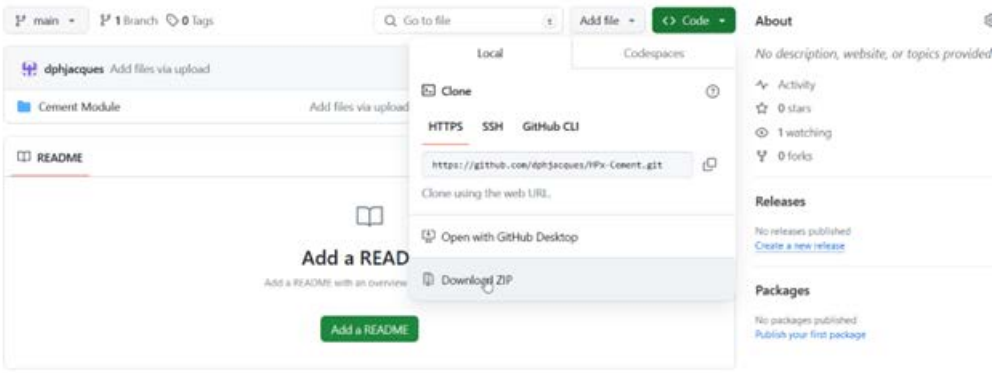
**You will learn to**

Add a module to HPGeochemistry

**Download Cement Module**

Go to the github repository via **link**

Download the repository via Code and Download ZIP




**Find the general folder for Modules**

The folders and files under Cement Module should be placed in the default Module folder of HPx/HYDRUS 5.x. It will then be available to be used it in different projects.

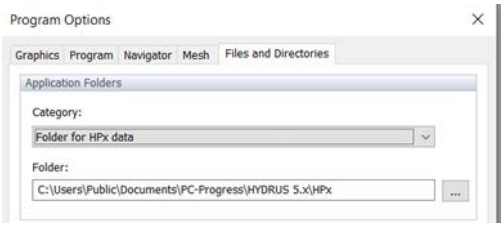
The default folder can be found via

(i) HYDRUS 5.0 GUI

Go to `Options->Program Options`...

Go to `Files and Directories` tab

Change the `Category` of the `Application Folders` to `Folder for HPx` data




The default folder is a subfolder of this folder. In this case, it is:

`C:\Users\Public\Documents\PC-Progress\HYDRUS 5.x\HPx\Modules`

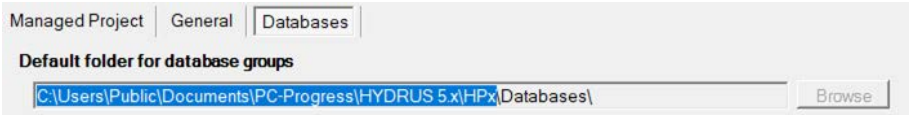
(ii) HPGeochemistry GUI

Go to `Settings and Preferences`

Go to the `Settings` tab

Go to the `Databases` subtab

The folder in the textbox "`Default folder for database groups`" indicates the default folder for databases. The parent folder gives also the path to subfolder `Modules`. Thus, in case you see




the default folder for modules is

`C:\Users\Public\Documents\PC-Progress\HYDRUS 5.x\HPx\Modules`

**Place the cement module in the general folder**

Make a subfolder `Cement Module` in the default folder for modules

Unzip the downloaded

Place the subfolder in the folder `HPx-Cement-main\Cement Module` in the subfolder `Cement Module`

*END TUTORIAL 3-1*

## 3.2 Templates

# 4. THERMODYNAMIC MODELLING OF CEMENT HYDRATES

## Chapter at a glance

# 4.1 Set-up HPGeochemistry

The tutorial for setting up a project, add databases and modules script is limited to a list of instructions. The reader is referred to XX for more details on the basic use of HPGeochemistry.

All building blocks are present in the downloaded database files and cement module.

*TUTORIAL 4-1 CREATE MANAGED PROJECT WITH THE CEMENT MODULE*

**Description**

Create the project for this tutorial and add the cement module in the project

**You will learn to**

To create a managed project

To add a module to the project

**Depends on**

Tutorial 3-1 Install CEMENT module

**Step 1 – Create project**

Click `Project Manager` button

Click `Create new project` button

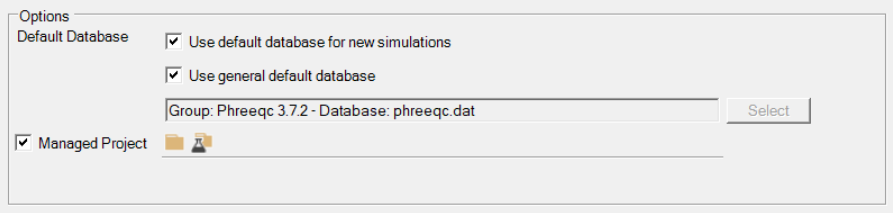
The `Metadata` dialogue window appears.

Give a name to your project

**Step 2 – Add cement module to the project**

A prerequisite to use a module in HPGeochemistry is that it is a managed project:

Ensure that the Managed Project is checked




For the Cement Module, folders for data, database and modules are required.

Open the `Folder Manager` for a managed project

Ensure that the folder `Data` (`04_data`) is checked

Options

Folders

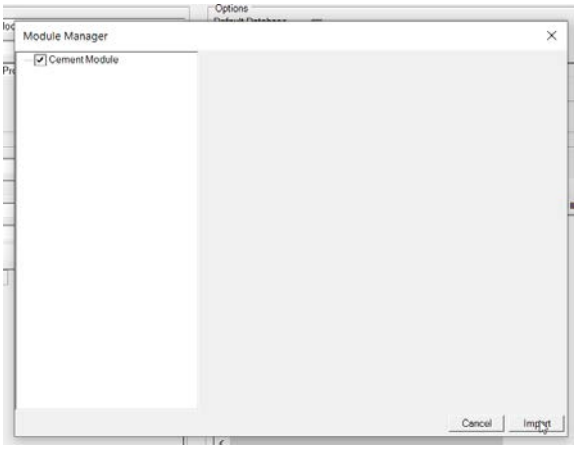
| Name | Folder | Create | Include | Files |
|---|---|---|---|---|
| Background Documents | 01_background-documents | ☑ | ☐ | ☑ |
| Literature Documents | 02_literature | ☑ | ☐ | ☑ |
| Model | 03_model-description | ☑ | ☐ | ☑ |
| Data | 04_data | ☑ | ☐ | ☑ |
| Database | 10_database | ☑ | ☑ | ☐ |
| Scripts | 11_scripts | ☑ | ☑ | ☐ |
| Modules | 12_modules | ☑ | ☑ | ☐ |
| GeoChemModels | 20_geochemical-models | ☑ | ☐ | ☐ |
| Transport | 21_transport-models | ☑ | ☐ | ☐ |
| Figures | 50_figures | ☐ | ☐ | ☑ |
| Report | 51_report | ☐ | ☐ | ☑ |
| Deprecated | 98_deprecated-files | ☐ | ☐ | ☑ |
| Temporary | 99_temporary-files | ☐ | ☐ | ☑ |

OK

Open the `Module Manager` for a managed project 

Check `Cement Module`

Click `Import`




Click **OK**

*END TUTORIAL 4-1*

# 4.2 Modelling CEM-I hydrate assemblage and pore water composition

## 4.2.1 Calculation of the hydrate assemblage for a CEM-I oxide composition

*TUTORIAL 4-2 CALCULATE THE CEMENT HYDRATES ASSEMBLAGE AND PORE WATER COMPOSITION FOR A CEM I*

**Description**

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

**You will learn to**

Start a simulation from a template in a module

**Depends on**

Tutorial 4-1 Create managed project with the cement module

**Step 1 – Create simulation from template for the calculation of the hydrates assemblages**

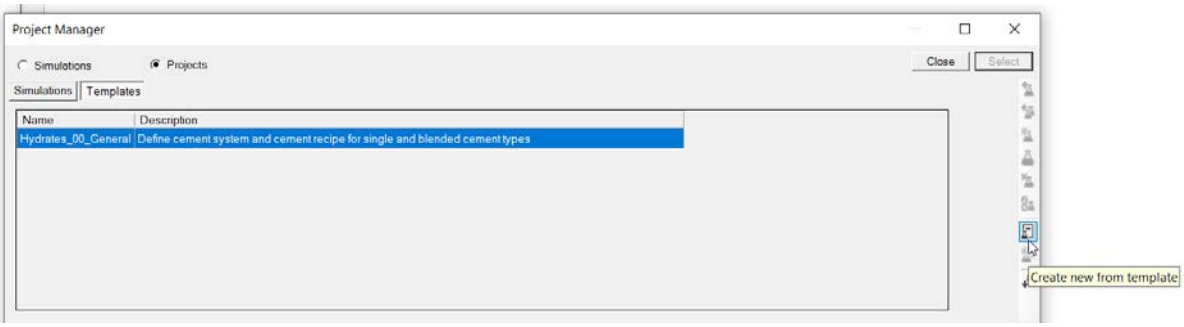
Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_00_general`"

Click the "`Create new from template`" button




The `Metadata` dialogue window opens.

Define the name of the simulation (`CEM I`)

Click `OK`

The Project Manager is active again.

Select the simulation (CEM I) and click `Select`

**Step 2 – Define the problem**

Only information in the input tables needs to be changed if values are different from the default values. All instructions and data are included in the simulation via the template.

Go to the `Global Definitions` input tab

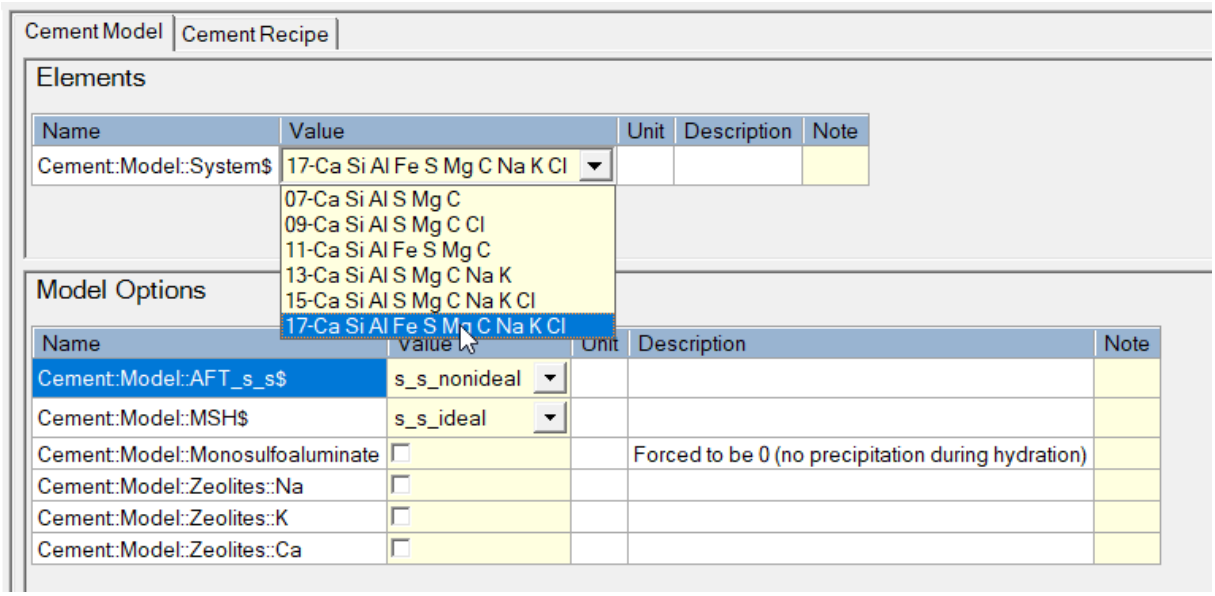
Go to the `Global Variables`

> The template consists of two input tables: "`Cement Model`" and "`Cement Recipe`" to, respectively, select the cement model and define the cement recipe.
>
> Go to the `Cement Model` tab

The first group "Elements" defines which elements are to be considered in the calculations. Selection is done via the drop down menu for the variable `Cement::Model::System$`.

> Select system 17 from the drop down menu. Note that this system also contains Cl but is neglected in this tutorial (no Cl in the clinker composition)

The second group "`Model Options`" allows to select some specific minerals to be included or excluded from the models – default selections are used in this tutorial.




> Go to the `Cement Recipe` tab

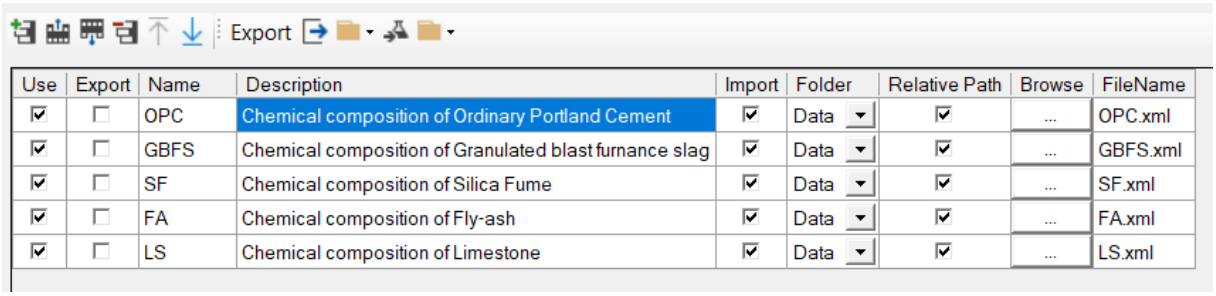
In the first group "Properties", the temperature (°C), water/cement ratio (-) and amount of cement (g) are given. In this tutorial, the default values are used.

The five other groups represent the ordinary Portland cement (OPC) clinker and following supplementary cementitious materials (SCMs): granulated blast furnace slag (GBFS), silica fume (SF), fly-ash (FA) and limestone (LS). For each material, it is possible to define oxide composition, the amount (mass%), the reacted amount (% reacted) and the bulk density (kg/dm$^3$).

> For the cement recipe in this tutorial (CEM-I),

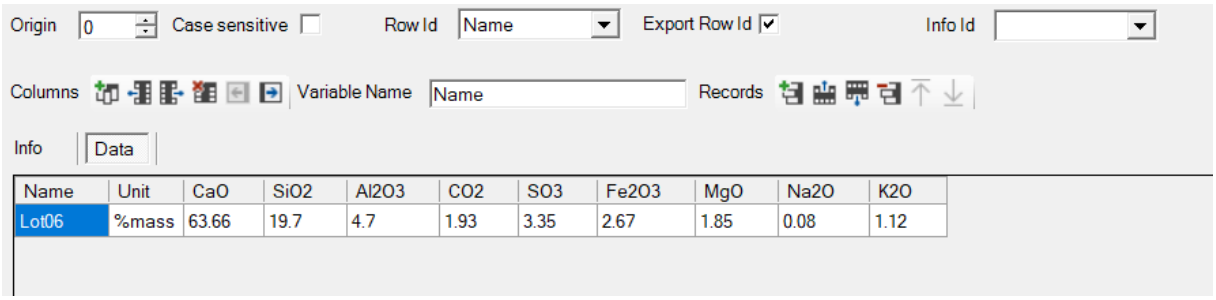
The composition of the OPC clinker and the SCMs are given via data tables defined in the DataBlock. Each record in the data table forms an item in the drop down list. Beside the default options, additional compositions can be defined (see Tutorial 4-4).

> Go to `DataBlock`



| Use | Export | Name | Description | Import | Folder | Relative Path | Browse | FileName |
|---|---|---|---|---|---|---|---|---|
| ☑ | ☐ | OPC | Chemical composition of Ordinary Portland Cement | ☑ | Data | ☑ | ... | OPC.xml |
| ☑ | ☐ | GBFS | Chemical composition of Granulated blast furnance slag | ☑ | Data | ☑ | ... | GBFS.xml |
| ☑ | ☐ | SF | Chemical composition of Silica Fume | ☑ | Data | ☑ | ... | SF.xml |
| ☑ | ☐ | FA | Chemical composition of Fly-ash | ☑ | Data | ☑ | ... | FA.xml |
| ☑ | ☐ | LS | Chemical composition of Limestone | ☑ | Data | ☑ | ... | LS.xml |

Select **OPC** data block in the top panel. Information in that data block is shown in the bottom panel:



Origin 0 Case sensitive Row Id Name Export Row Id Info Id

Columns Variable Name Name Records

Info Data

| Name | Unit | CaO | SiO2 | Al2O3 | CO2 | SO3 | Fe2O3 | MgO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|---|---|
| Lot06 | %mass | 63.66 | 19.7 | 4.7 | 1.93 | 3.35 | 2.67 | 1.85 | 0.08 | 1.12 |

For each of the cement constituents (OPC and SCMs), a data table is available. In each of these data table, an oxide composition is given by default (see example for OPC). The indicator in the column `Name` is used in the drop down list. The reference to the oxide composition is listed in the `Info` tab. Other compositions can be added (see Tutorial 4-4).

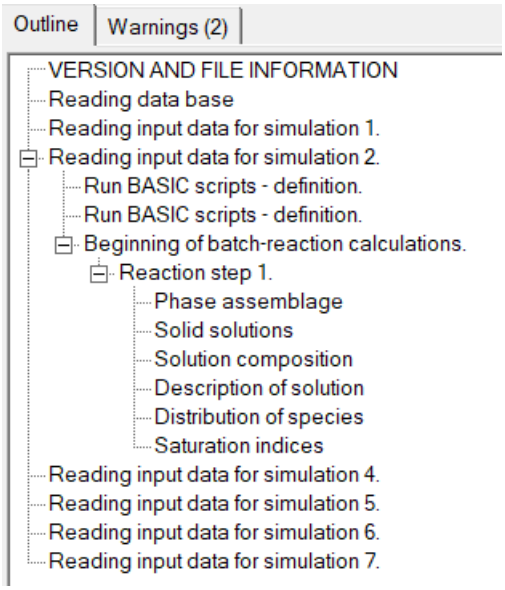
Run the problem 

**Step 3 – Check the output**

Go to `phreeqc.out` file

Go to `Outline` tab

Go to `Reading input data for simulation 2` -> `Beginning of batch-reaction calculation`



Outline | Warnings (2)

- VERSION AND FILE INFORMATION
- Reading data base
- Reading input data for simulation 1.
- Reading input data for simulation 2.
  - Run BASIC scripts - definition.
  - Run BASIC scripts - definition.
  - Beginning of batch-reaction calculations.
    - Reaction step 1.
      - Phase assemblage
      - Solid solutions
      - Solution composition
      - Description of solution
      - Distribution of species
      - Saturation indices
- Reading input data for simulation 4.
- Reading input data for simulation 5.
- Reading input data for simulation 6.
- Reading input data for simulation 7.

Note first that there are warnings. This is because Cl-containing hydrates are in the hydrate assemblage (due to the selection of system 17), but there is no Cl in the input.

The output file gives the standard phreeqc output related to a batch reaction calculation. All geochemical information can be obtained from this file. The template also produces a special output file with the most important information for the cement system.

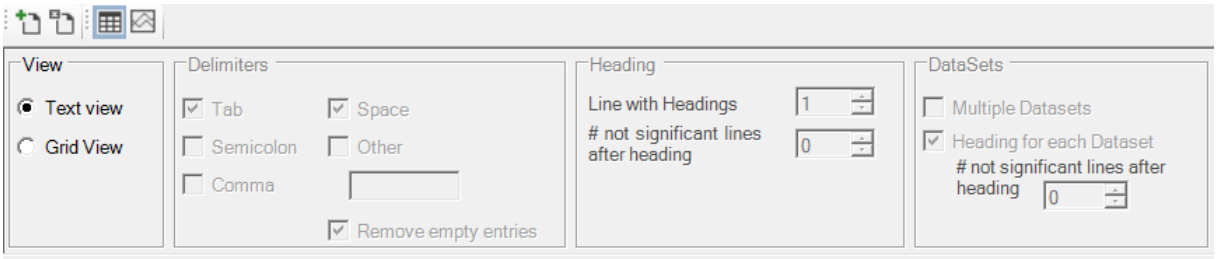
Go to the Data File output workspace

Click Open Data File

Open the file "Cement_Hydration_1001.out"

Go to the View options

Select Text View




Cement_Hydration_1001.out

```
CEMENT RECIPE
                        OPC clinkers            100 weight%
                         Reacted OPC            100 % reacted

                    Amount of cement            100
                           w/c ratio           0.50
                         Temperature             25

CONTENT
              Name          Moles
               CaO       1.14e+00
               SiO2      3.28e-01
              Al2O3      4.61e-02
               SO3       4.18e-02
               CO2       4.39e-02
               MgO       4.59e-02
              Fe2O3      1.67e-02
               Na2O      1.29e-03
               K2O       1.19e-02

HYDRATED PHASES

              Name          Moles          Grams          Volume

       Portlandite       4.17e-01          30.90           13.79
               Cal       3.49e-02           3.50            1.29
     hydrotalcite       1.15e-02           5.09            2.53
   monocarbonate       5.04e-03           2.87            1.32
               AFT       4.56e-02          18.95           10.68
               CSH       4.28e-01          61.52           27.22
    HydrogarnetFe       3.11e-02          13.35            4.53
               MSH       9.51e-16           0.00            0.00
```

In this view, you find information on the cement recipe, the oxide composition, the hydrated phases and the pore water composition, calculated volumes, and information on solid solution composition.

*END TUTORIAL 4-2*

## 4.2.2 Limited replacement OPC with GBFS

*TUTORIAL 4-3 CALCULATE THE CEMENT HYDRATES ASSEMBLAGES WHEN PART OF OPC IS REPLACED BY GBFS*

**Description**

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C where 10% of OPC is replaced by GFBS. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1); the GBFS oxide composition from Adu-Amankwah et al. (2017). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

**You will learn to**

Define different cases in one simulation

**Depends on**

Tutorial 4-2 Calculate the cement hydrates assemblage and pore water composition for a CEM I

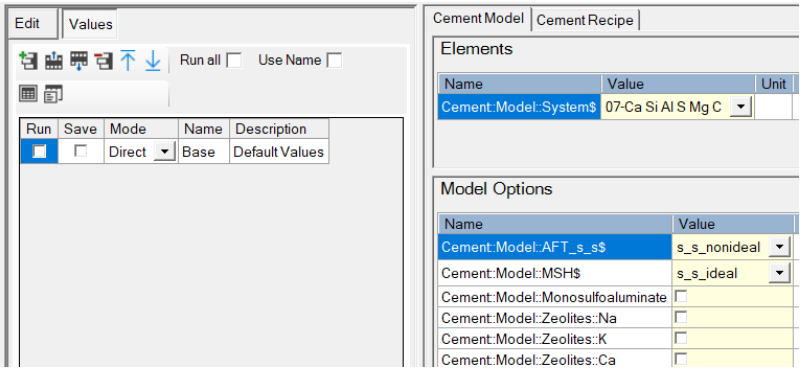
**Step 1 – Create the case(s) in the simulation**

Open the simulation that was created in Tutorial 4-2 via the Project Manager or via the drop down list of previously opened simulations.

Go to the Global Variable Tab

Go to Global Variables

Cases are listed in the left panel




In a re-opened simulation, the table contains one case called Base by default. In a newly made simulation, this table is still empty, and cases can be added by the procedure described below.

Ensure that Use Name is checked. The subfolders containing the cases will have the name printed in the column Name.

Click to add a case – and add a case respectively before or after the selected case.

Ensure that you have two cases

Call the first case "CEMI" and check the box in the column with heading Save

Call the first case "CEMI_variant01" and check the box in the column with heading Save.

The first case uses the default values in the table, i.e. 100% OPC.

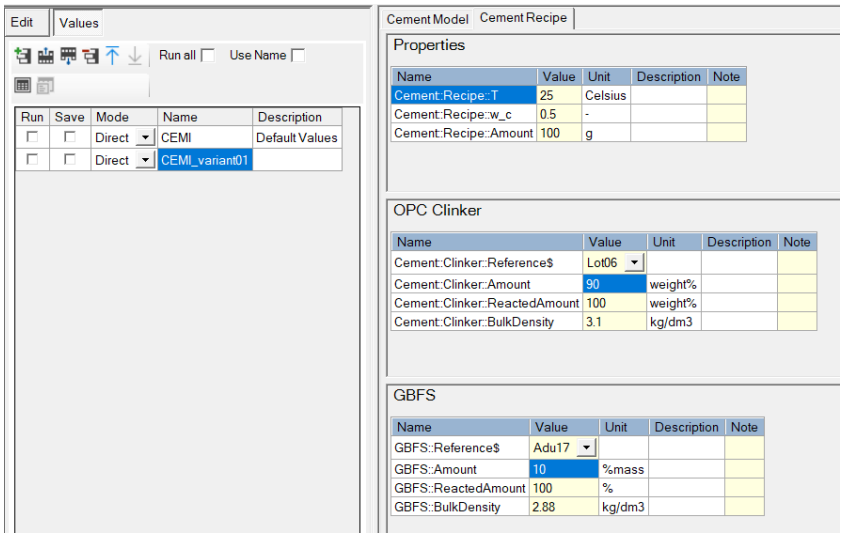
Select the second case (just select any cell in the second row)

Go to the Cement Recipe tab

Change following variables:

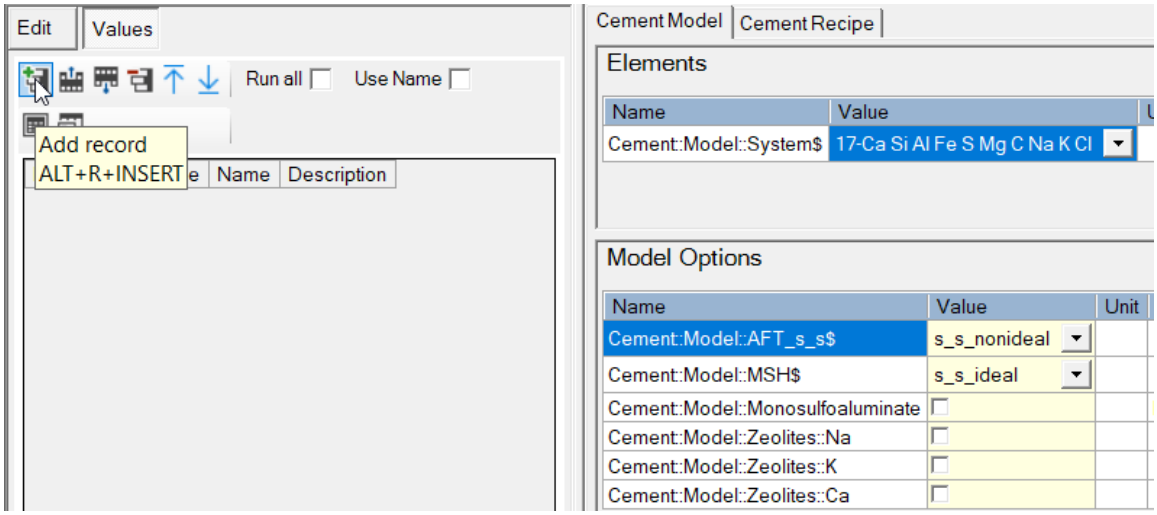
Cement::Clinker::Amount to 92

GBFS::Amount to 8




can be added via the left panel of the Global Variables Table

Click the icon to add a record to the




*END*

*TUTORIAL 4-4 CHANGE THE CLINKER OXIDE COMPOSITION OF OPC*

**Description**

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

**You will learn to**

Start a simulation from a template in a module

**Depends on**

Tutorial 4-1 Create managed project with the cement module

**Step 1 – Create simulation from template for the calculation of the hydrates assemblages**

G

Cement Model | Cement Recipe

Properties

| Name | Value | Unit | Description | Note |
|---|---|---|---|---|
| Cement::Recipe::T | 25 | Celsius | | |
| Cement::Recipe::w_c | 0.5 | - | | |
| Cement::Recipe::Amount | 100 | g | | |

OPC Clinker

| Name | Value | Unit | Description | Note |
|---|---|---|---|---|
| Cement::Clinker::Reference$ | Lot06 | | | |
| Cement::Clinker::Amount | 100 | weight% | | |
| Cement::Clinker::ReactedAmount | 100 | weight% | | |
| Cement::Clinker::BulkDensity | 3.1 | kg/dm3 | | |

GBFS

| Name | Value | Unit | Description | Note |
|---|---|---|---|---|
| GBFS::Reference$ | Adu17 | | | |
| GBFS::Amount | 0 | %mass | | |
| GBFS::ReactedAmount | 100 | % | | |
| GBFS::BulkDensity | 2.88 | kg/dm3 | | |

# 4.3 Modelling CEM-II and CEM-III hydrated phases and pore water composition

## 4.3.1 Reaction path model with replacements

*TUTORIAL 4-5 BLENDING OPC AND GBFS ALONG A REACTION PATH*

**Description**

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C when OPC is replace by GBFS up to 60%. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1); the GBFS oxide composition from Adu-Amankwah et al. (2017). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

**You will learn to**

Include a parametric study

**Depends on**

Tutorial 4-1 Create managed project with the cement module

**Step 1 – Create the simulation from the template to blend materials**

Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_10_Bleding_general`"

Click the "`Create new from template`" button

The `Metadata` dialogue window opens.

Define the name of the simulation (`Blending`)

Click `OK`

The Project Manager is active again.

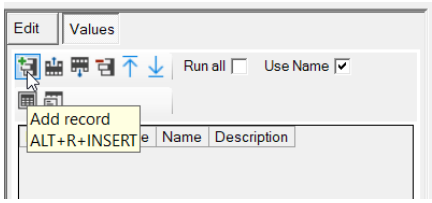
Select the simulation (Blending) and click `Select`

**Step 2 – Create the reaction path of blending**

Go to `Global Definitions`

Go to `Global Variables`

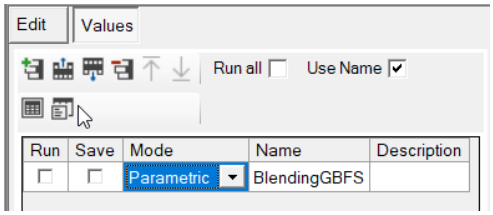
Add a case




Give the case a name (`BlendingGBFS`)

Select the `Parametric` Mode

Click on `Select Mode` ( ) icon

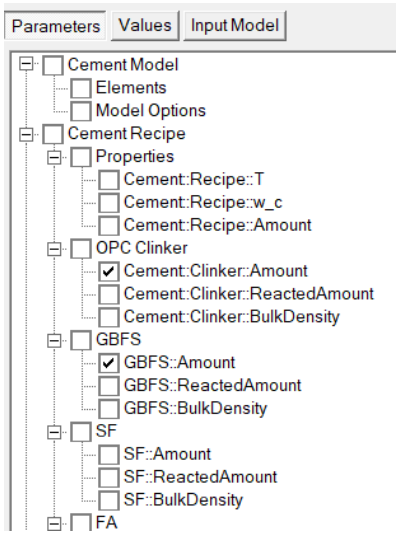



At the right side, the panel to define a parametric study is visible. There are three panels.

The first panel serves to select the variables included in the parametric study. It consists of all numerical variables in the global input table; Only selection at the third level are relevant.

Select `Cement::Clinker::Amount`

Select `GBFS::Amount`



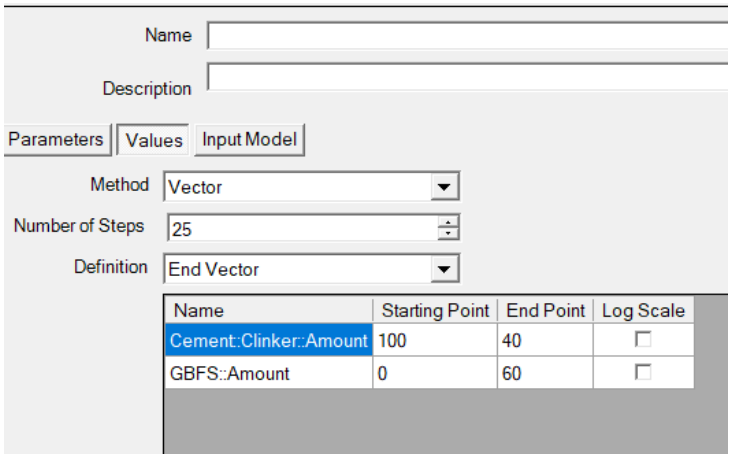

Go to the second panel (`Values`)

In this panel, the method (to obtain different parameter values) and the options relevant to the method are available.

Select method `Vector`.

In the Vector method, one defines the start and end point of the vector (or the start point and the increment), the number of increments, and the corresponding numerical values. The vectors with variable values are calculated equally between the start and end point.

Fill in the number of increments (e.g. 25)




The third panel identifies the part of the code that needs to be executed with different parameter values. One option is to run the complete input file – PHREEQC will be restarted every time. Output files will be overwritten.

A better way is to put the relevant code in a separate file:

Select Specific File

Select `Modules` from the drop down box `Folder`

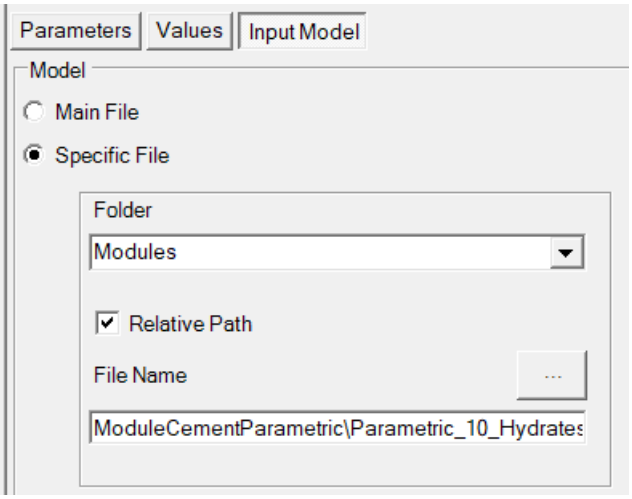
Check `Relative Path`

Click …

Go to `ModuleCementParametric` Folder

Select `Parametirc_10_Hydrates.hps`

Click `Open`




**Step 3 – Set Model and Output Options**

Go to the input table via the icon 

Choose model 17-Ca Si Al/fe S Mg C Na K Cl in the `Cement Model` tab – `Elements` group

Go to the `Output` Tab

Details of the output tab is given in 9.1.1 (not available yet).

The first group (`Output – Values`) allows to select the output variables. Hydrates can be reported in moles, volumes ($cm^3$) and weight (g). Also the saturation index of the hydrates can be send to the output.

The second group (`Output – Figures – General`) allows the select the general settings of the figures. The crucial records are:

`Cement::Figures::Lead$` - define the output variable used for the X-axis. This can be one of the (numerical) variables in the global input table. If _null_, integer values will be used.

Type `GBFS::Amount` in `Cement::Figures::Lead$`

`Cement::Figures::X::Title$` - define the label of the X-axis

Type `GBFS [%]` in `Cement::Figure::X::Title$`

The third group (Output – Figures – Plots) allows to select the type of plots:

`Cement::Figures::Water::Elements::Y::Scale$` - to express concentrations in M, mg/L or both – select both

`Cement::Figures::Solids::Detail$` - not implemented yet – do not change

`Cement::Figures::Solids::Units$` - to express the hydrates for the total system or per 100 g – select both

`Cement::Figures::Solids::Value$` - to define the unit of the hydrates (moles, volume, weight) – select All

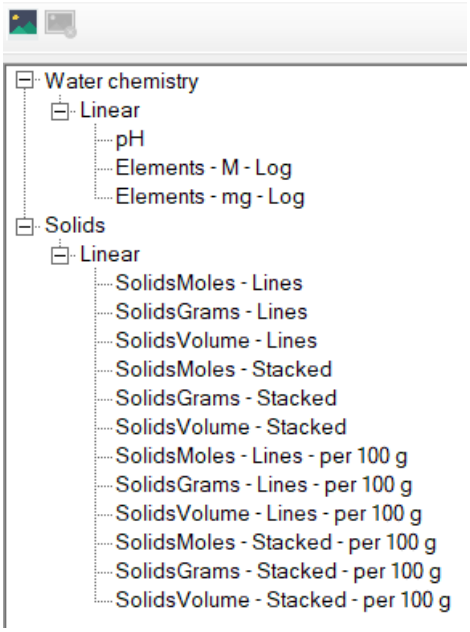
`Cement::Figures::Solids::Type$` - to define the appearance of the plot (lines or stacked lines) – select Both

Run 

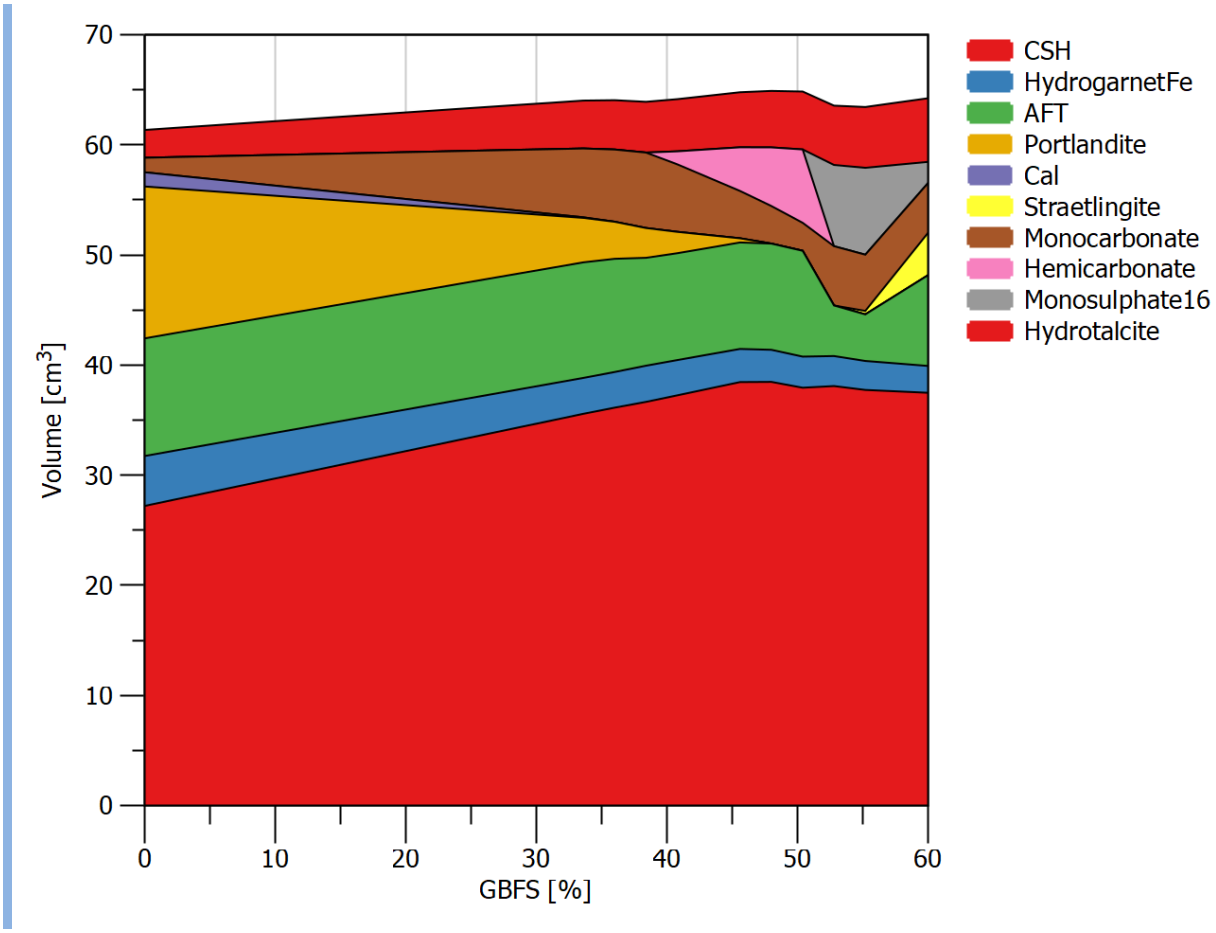
**Step 4 – Check Plots**

Go to the `Graph Work` Space 

By selecting plots on the list of plots, simulation results can be viewed. The selected options give a range of different plots.




Below is the stacked curve of the volumes of the hydrates as a function of % GBFS. *Need to be done: add initial volume (volume clinkers+volume water), capillary water, gel water, chemical shrinkage. Colors need to be optimized…*


70
60
50
40
30
20
10
0
Volume [cm3]
0
10
20
30
40
50
60
GBFS [%]
CSH
HydrogarnetFe
AFT
Portlandite
Cal
Straetlingite
Monocarbonate
Hemicarbonate
Monosulphate16
Hydrotalcite

# 6. MODELLING DEGRADATION PROCESSES – THERMODYNAMIC APPROACHES (MIXING TANK APPROACHES)

## Chapter at a glance

# 6.1 Ingress

*TUTORIAL 6-1 CARBONATION OF CEM-I*

**Description**

Simulate the carbonation of CEM-I (100% OPC) of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al-S-Mg-C system.

**You will learn to**

**Depends on**

Tutorial 4-1 Create managed project with the cement module

**Step 1 – Create the simulation from the template to blend materials**

Go to `Project Manager` 

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_20_Ingress_general`"

Click the "`Create new from template`" button 

The `Metadata` dialogue window opens.

Define the name of the simulation (`Carbonation`)

Click `OK`

The Project Manager is active again.

Select the simulation (`Carbonation`) and click `Select`

**Step 2 – Define the carbonation process**

Go to `Global Definitions`

Go to `Global Variables`

Cement model and cement recipe are the default values

Go to `Ingress` tab

In the first group (Type of Input), the type of ingress can be defined (only Components is implemented). With the selection Components, the components in the group Define Component will be added to the system. The total amount of added components is defined in absolute terms (not implemented) or relative terms to one of the oxides in the cement system. In this particular case, CO2 will be added to 0.99% of the amount of CaO (more will lead to non-convergence) in 101 steps.

`Cement::Ingress::Amount` 0.99

Cement Model | Cement Recipe | Ingress | Output

Type of Input

| Name | Value | Unit |
|---|---|---|
| Cement:Ingress::Type$ | Components | |
| Cement:Ingress::Amount:Type$ | Relative | |
| Cement:Ingress::Amount:Component$ | CaO | |
| Cement:Ingress::Amount | 0.99 | |
| Cement:Ingress::Steps | 101 | |

In the second group (Define Components), the relative amount of a component is defined – here the relative amount of $CO_2$ per mol of CaO is defined.

`Cement::Ingress::Components::CO2` 1

Define Components

| Name | Value |
|---|---|
| Cement:Ingress::Components::Type$ | Relative Moles |
| Cement:Ingress::Components::CO2 | 1 |
| Cement:Ingress::Components::SO3 | 0 |
| Cement:Ingress::Components::MgCl2 | 0 |
| Cement:Ingress::Components::NaCl | 0 |

**Step 3 – Plots**

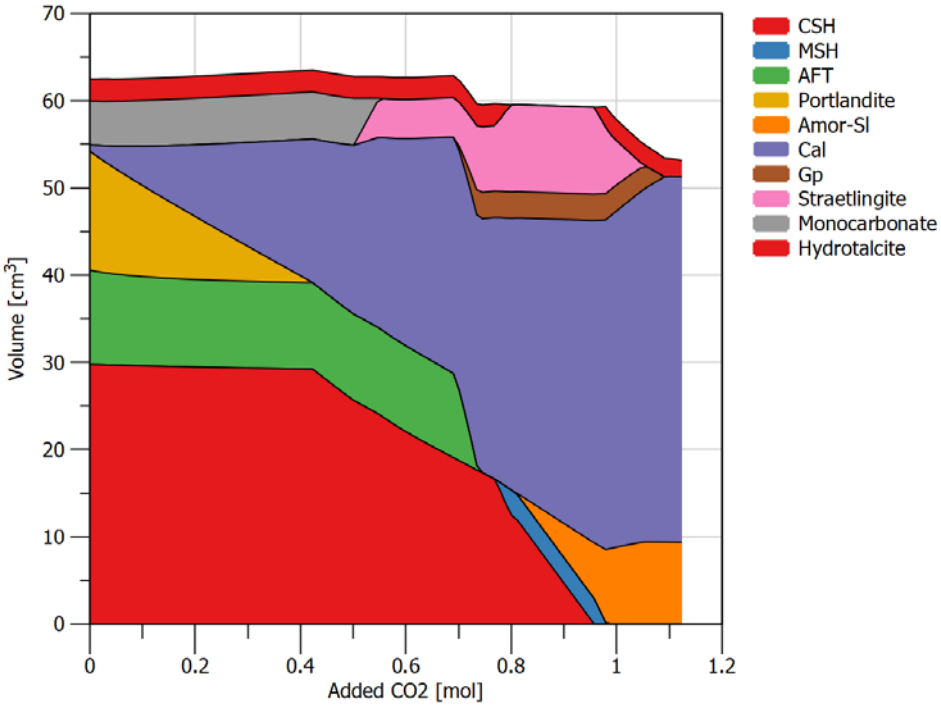
Go to the `Output` tab and select the desired plots

Change `Cement::Figures::Lead$` to `Cement::Ingress::Components::CO2` to have the lead variable (X-axes) the absolute amount of $CO_2$ that have been added.

`Change Cement::Figures::X::Title$` to `Added CO2 [mol]`

Run

Example figure: Stacked plot of solids expressed in volumes



## 6.3 Leaching

*TUTORIAL 6-2 LEACHING OF CEM-I WITH A SOLUTION AT PH 3*

**Description**

Simulate the leaching of CEM-I (100% OPC) of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The leaching solution has a pH of 3 (bufferd by HCl) – assume a cumulative leaching to 500 L of water. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al-S-Mg-C-Na-K system.

**You will learn to**

**Depends on**

Tutorial 4-1 Create managed project with the cement module

**Step 1 – Create the simulation from the template to blend materials**

Go to `Project Manager` 

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_30_Leaching_general`"

Click the "`Create new from template`" button 

The `Metadata` dialogue window opens.

Define the name of the simulation (`Leaching`)

Click `OK`

The Project Manager is active again.

Select the simulation (`Leaching`) and click `Select`

**Step 2 – Define the carbonation process**

Go to `Global Definitions`

Go to `Global Variables`

Go to `Cement Model` tab

`Cement::Model::System$` change to system 13

Change the AFt non-ideal model to ideal model (otherwise, PHREEQC does not convergence when AFt is almost completely dissolved):

`Cement::Model::AFT_s_s$` change to `s_s_ideal`

Go to `Leaching` tab

In the first group (Leaching amounts), the cumulative amount of leaching is defined in four steps of respectively increments of 0.01, 0.1, 1 and 10 L. The default values leach first 1 L with 0.01 L increments, 9 L with 0.1 L and 90 L with 1 L, in total 100 L thus.

Adapt the fourth step (10 L) to have a total amount of 500 L

`Cement::Leaching::Step` 400

The `cement::Leaching::Type$` is selected on `Components`. The composition of the leaching solution is defined in terms of concentration of different components given in the second group

`Components`. Default values are 1E-15M – this will avoid to have warnings during calculations linked to the presence of phases in the model without the corresponding elements in solution.

`Change Cement::Leaching::Cl` to 1E-3

`Change Cement::Leaching::pH` to 3

Select `Cl` in the drop down list `Cement::Leaching::Charge$`

The solution thus have a pH of 3 and Cl concentration is adapted to have a charge-balanced solution. This will result in following PHREEQC input during calculations (for increment of 0.01L):

```
solution 169898
-water 0.01
-units mol/kgw
Ca 1e-15
Al 1e-15
pH 3
Cl 0.001 charge
C 1e-15
K 1e-15
S 1e-15
Na 1e-15
Mg 1e-15
Si 1e-15
```

**Step 3 – Plots**

Go to the `Output` tab and select the desired plots

Group `Output – Figures - General`

Change `Cement::Figures::X::Title$` to `Added water [kg]`

Change `Cement::Figures::X::Scale$` to `Logscale` or `Both`

Change `Cement::Figures::Y::Grid$` to `Both Log`

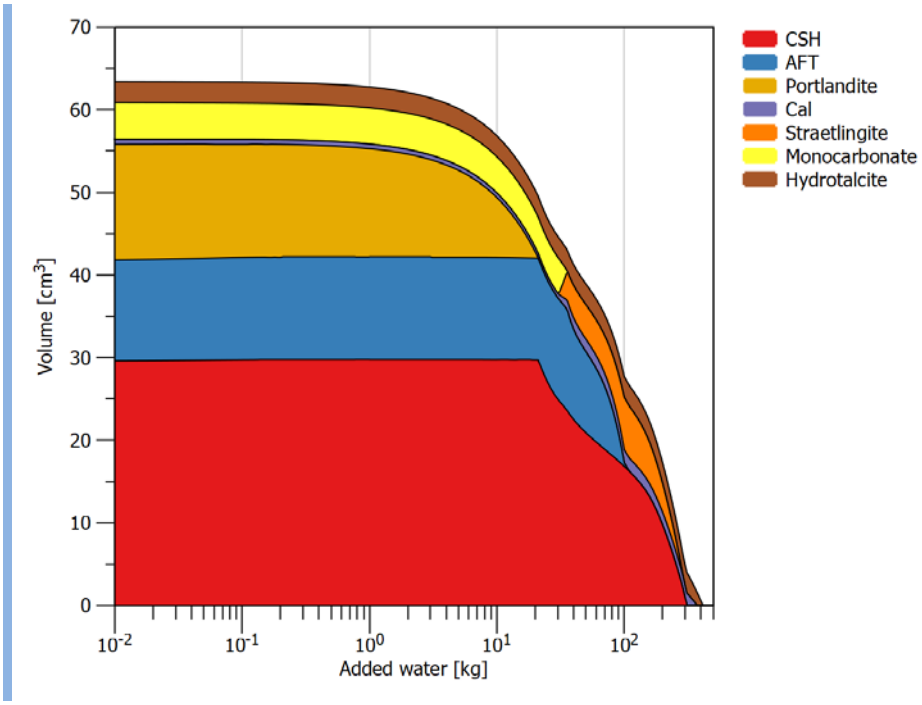
Group `Output – Figures – Plots`

Change `Cement::Figures::Solids::Value$` to `All`

Change `Cement::Figures::Solids ::Types$` to `Both`

Run

Example figure: Stacked plot of solids expressed in volumes


70
60
50
40
30
20
10
0
Volume [cm3]
10-2
10-1
100
101
102
Added water [kg]
CSH
AFT
Portlandite
Cal
Straetlingite
Monocarbonate
Hydrotalcite

# INDEX


## C

Case, 9, 14

## D

Define a Case, 14

## P

Parametric study, 9